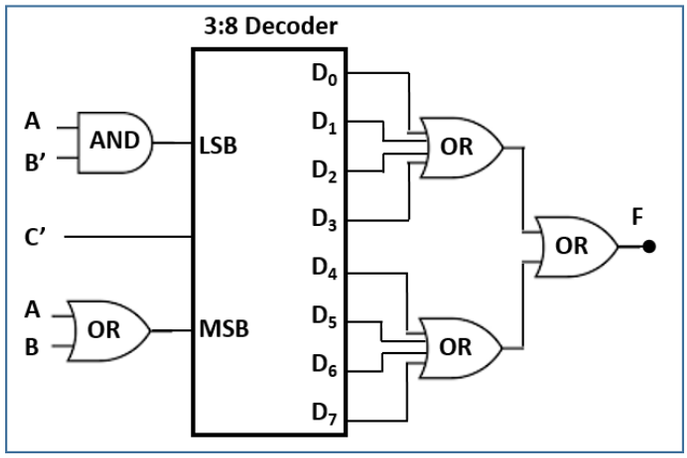
Started on	Friday, 21 May 2021, 10:04 AM
State	Finished
Completed on	Friday, 21 May 2021, 11:24 AM
Time taken	1 hour 19 mins
Marks	23.00/26.00
Grade	**17.69** out of 20.00 (**88**%)

Question **1**

Correct

Mark 1.00 out of 1.00

A circuit, as shown below, was implemented using a 3:8 Decoder, AND, and OR gates to generate the output function F. Which of the following expression represents the fully reduced form of F?



Select one:

- 1 ✔
- 0
- $AB' + C'$
- $AB + C'$
- $A + B + C'$
- $A + B' + C'$

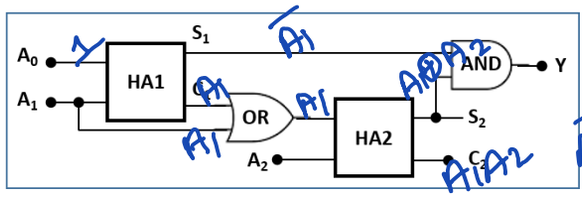
Your answer is correct.

The correct answer is: 1

Question **2**

Incorrect

Mark 0.00 out of 1.00



A circuit, as shown above, is implemented using two half-adders, an OR and an AND gate. Which of the following statement is true for the output Y?

Select one:

- For $A_0 = 1, Y = A_1 A_2$
- For $A_0 = 0, Y = A_1 A_2$ ✖
- For $A_0 = 0,\ Y = A_1^{'} A_2$
- For $A_0 = 1\ Y = A_1^{'} A_2$

Your answer is incorrect.

The correct answer is: For $A_0 = 1\ Y = A_1^{'} A_2$

Question **3**

Incorrect

Mark 0.00 out of 1.00

Identity I_1 : a XOR b = a NOR b

Identity I_2 : a XOR b = NOT(a NOR b)

Identity I_3 : a XOR b XOR c =a NOR b NOR c

Identity I_4 : a XOR b XOR c = NOT(a NOR b NOR c)

Which of the above Identities is correct?

Select one:

- I_2 and I_3
- I_2 and I_4 ✖
- I_1 and I_4
- only I_2

Your answer is incorrect.

The correct answer is: I_2 and I_3

Question **4**

Correct

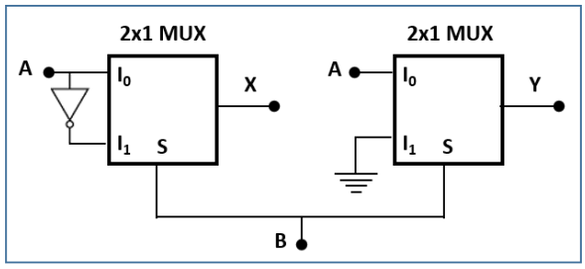
Mark 1.00 out of 1.00

A circuit is implemented using two 2:1 multiplexers as shown.

A and B are inputs, X and Y are outputs.

The ground represents logic 0.

Which of the following is true?



Select one:

- The circuit works as a half-subtractor (B - A) with Y = Difference; X = Borrow
- The circuit works as a half-subtractor (A - B) with X = Difference; Y = Borrow
- The circuit works as a half-subtractor (A - B) with Y = Difference; X = Borrow
- The circuit works as a half-subtractor (B - A) with X = Difference; Y = Borrow ✔

Your answer is correct.

The correct answer is: The circuit works as a half-subtractor (B - A) with X = Difference; Y = Borrow

Question **5**

Correct

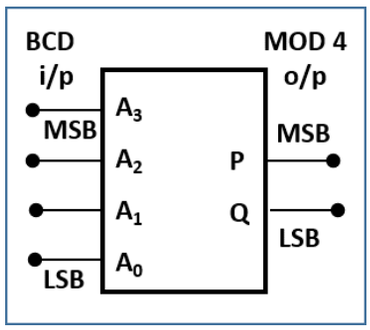
Mark 1.00 out of 1.00

A digital device takes only a valid 8421 BCD (i.e., from decimal equivalent 0 to 9) as input and returns Modulo 4. For any invalid input, the output is zero. The block diagram of the device is shown below. The fully reduced boolean expressions of P and Q in terms of A_3, A_2, A_1 and A_0 are _____

Note:-

$4 Modulo 4 = 0, 5 Modulo 4 = 1, 6 Modu$

The decimal equivalent of the input is divided by 4 and the remainder is the output.



Select one:

- $P = A_3' A_1$ and $Q = A_3' A_0 + A_2' A_1' A_0$ ✔
- $P = A_3' A_1$ and $Q = A_3' A_0 + A_2 A_1' A_0$
- $P = A_3 A_1$ and $Q = A_3' A_0 + A_2' A_1' A_0$
- $P = A_3' A_1$ and $Q = A_3' A_0$

Your answer is correct.

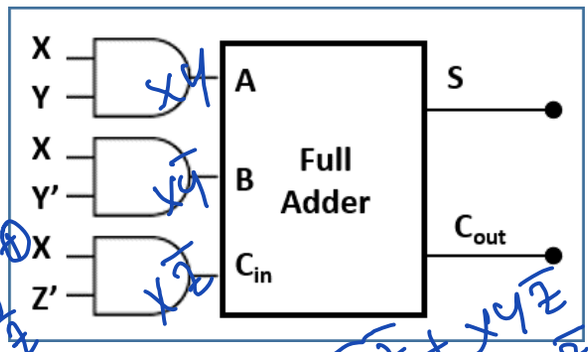
The correct answer is: $P = A_3' A_1$ and $Q = A_3' A_0 + A_2' A_1' A_0$

Question **6**

Correct

Mark 1.00 out of 1.00

For the full adder circuit shown below, which of the following expression for the sum (S) and carry-out (C_{out}) is correct?



Select one:

- $S = X + X'Z$
 $C_{out} = XZ'$
- $S = X + X'Z$
 $C_{out} = XZ$
- $S = XZ$
 $C_{out} = XZ'$
- $S = XZ'$
 $C_{out} = XZ$

Your answer is correct.

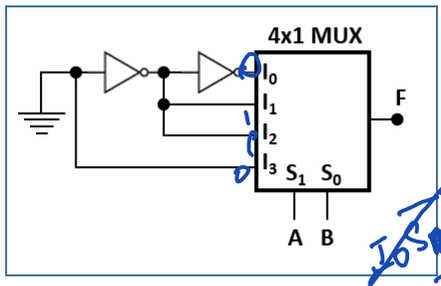
The correct answer is: $S = XZ$
 $C_{out} = XZ'$

Question **7**

Correct

Mark 1.00 out of 1.00

For the 4x1 multiplexer circuit shown below, A and B are the inputs to the select lines, and F is the output. The ground represents logic 0. Which of the following expression for F is correct?



Select one:

- $F = A' \ xor \ B$
- $F = A \ xor \ B'$
- $F = A \ xnor \ B$
- $F = A \ xor \ B$ ✔

Your answer is correct.

The correct answer is: $F = A \ xor \ B$

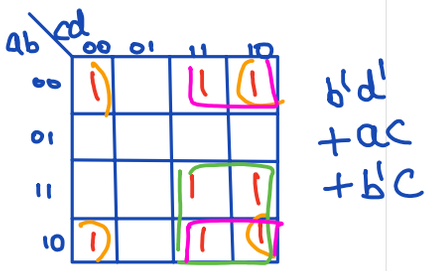
Question **8**

Correct

Mark 1.00 out of 1.00

$$F(a,b,c,d) = a'b + c'd + abc'd'$$

Reduced boolean expression for complement of F given by F' is:



Select one:

- $bd' + b'c + ac$
- $b'd' + bc + ac$
- $b'd' + b'c + ac$ ✔
- $b'd' + b'c + a'c$

Your answer is correct.

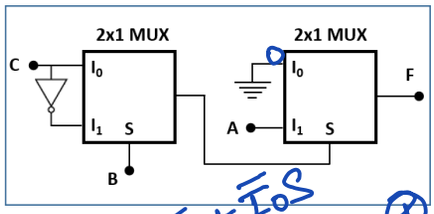
The correct answer is: $b'd' + b'c + ac$

Question **9**

Correct

Mark 1.00 out of 1.00

The circuit shown below is implemented using 2x1 multiplexers with I_0 and I_1 as inputs and S as select line. The ground represents logic 0. Which of the following expressions for $F(A,B,C)$ is correct?



$I_0\bar{S}+\bar{I}_0S$

$A(C \oplus B)$

Select one:

- $F=A'(B\ xor\ C)$
- $F=A(B\ xor\ C)$ ✔
- $F=A'(B\ xnor\ C)$
- $F=A(B\ xnor\ C)$

Your answer is correct.

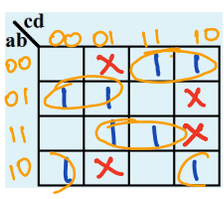
The correct answer is: $F=A(B\ xor\ C)$

Question **10**

Correct

Mark 1.00 out of 1.00

A function $F(a,b,c,d)=[?][?]\Sigma(2,3,4,5,8,10,13,15)$ and a don't care $d(a,b,c,d)=\Sigma(1,6,9,14)$ are populated in the K-map given below.



Which of the following represents the fully reduced form of $F[?][?]$?

$a'bc'+a'b'c+abd+ab'd'$

Select one:

- $a'b'c'+a'bc+abd+ab'd'$
- $a'b'c+a'bc'+abd+ab'd'$ ✔
- $c'd+cd'+a'b'c+a'bc'$
- $a'b'c+a'bc'+abd'+ab'd$

Your answer is correct.

The correct answer is:
$a'b'c+a'bc'+abd+ab'd'$

Question 11
Correct
Mark 1.00 out of 1.00

Simplify the following using De Morgan's theorem:

$\overline{\overline{a} + b \ . \ c}$

Note: bar represents the NOT operation, for eg. $\overline{X} = X'$

Select one:

- $a'+b+c'$ ✔
- $a+b'+c$
- $a'b+c'$
- $a'+bc'$

Your answer is correct.

The correct answer is: $a'+b+c'$

Question 12
Correct
Mark 1.00 out of 1.00

Reduce the boolean expression: $a'b+ab'+ab$

Select one:

- $a'+b$
- $a+b'$
- $a'b'$
- $a+b$ ✔

Your answer is correct.

The correct answer is: $a+b$

Question **13**

Correct

Mark 1.00 out of 1.00

$F=[(a'+d').(a'+c').(c+d').(b+c'+d)$

is populated in the K-map given below.

ab \ cd	00	01	11	10
00	0	1	3	1
01	4	1	7	6
11	12	1	1	1
10	8	1	1	1

Choose the correct minterm representation.

Select one:

- $F(a,b,c,d)=[?][?]\Sigma(1,2,5,9,10,11,13,$ ✔
- $F(a,b,c,d)=[?][?]\Sigma(0,2,4,6,7,8,12)$
- $F(a,b,c,d)=[?][?]\Sigma(1,3,5,9,10,11,13,$
- $F(a,b,c,d)=[?][?]\Sigma(0,3,4,6,7,8,12)$

Your answer is correct.

The correct answer is:

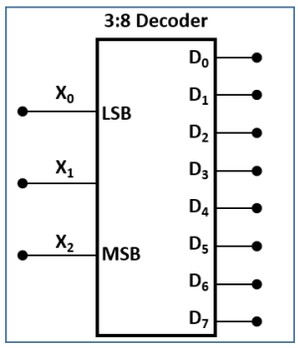
$F(a,b,c,d)=[?][?]\Sigma(1,2,5,9,10,11,13,14$

Question **14**

Correct

Mark 1.00 out of 1.00

A 3:8 decoder has a defect. For any $X_2X_1X_0$, each of the pins D_1, D_4, D_5 give output 0 and never give 1



$\bar{X_1} = 0$

A possible fault in the decoder may be _____

Select one:

- Pin X_1 is stuck-at-0
- Pin X_0 is stuck-at-1
- Pin X_0 is stuck-at-0
- Pin X_1 is stuck-at-1 ✔

Your answer is correct.

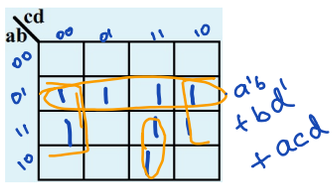
The correct answer is: Pin X_1 is stuck-at-1

Question **15**

Correct

Mark 1.00 out of 1.00

A function $F(a,b,c,d) = [?][?]\Sigma(4,5,6,7,11,12,14,15$ is populated in the K-map given below.



Which of the following represents the fully reduced form of F?

Select one:

- $a'b+bd'+acd$ ✔
- $ab+bd'+acd$
- $a'b+bd+acd$
- $a'b+bd'+acd'$

Your answer is correct.

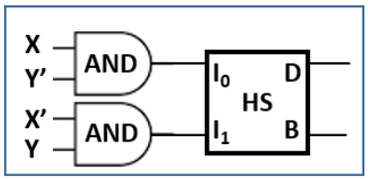
The correct answer is: $a'b+bd'+acd$

Question 16

Correct

Mark 1.00 out of 1.00

A half-subtractor (HS) shown below takes I_0 and I_1 as inputs and gives ($I_1 - I_0$) as output. Which of the following expressions for the difference D and borrow B is correct?



Select one:

- $D = X \ xor \ Y$ and $B = X'Y$
- $D = X' \ xor \ Y$ and $B = XY'$
- $D = X \ xor \ Y$ and $B = 1$
- $D = X' \ xor \ Y$ and $B = X'Y$
- $D = X \ xor \ Y$ and $B = XY'$ ✔

Your answer is correct.

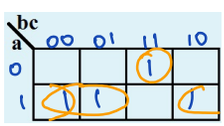
The correct answer is: $D = X \ xor \ Y$ and $B = XY'$

Question 17

Correct

Mark 1.00 out of 1.00

A function $F(a,b,c) = [?][?]\Sigma(3,4,5,6)$ is populated in the K-map given below.



a'bc + ab' + ac'

Which of the following represents the fully reduced form of $F[?][?]$?

Select one:

- $ab + ac' + a'b'c[?][?]$
- $ab' + ac + a'bc'$
- $ab' + ac' + a'bc$ ✔
- $ab' + ac' + a'b'c$

Your answer is correct.

The correct answer is: $ab' + ac' + a'bc$

Question 18

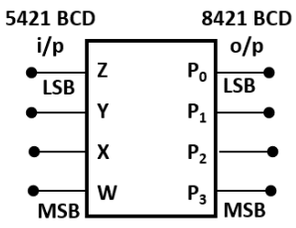
Correct

Mark 1.00 out of 1.00

An IC converts input 5421 BCD code as shown below in table to output 8421 BCD. If invalid inputs do not affect the output of the converter (don't care), which of the following is the correct expression for the LSB output bit P_0?

5421 BCD

Decimal	W (5)	X (4)	Y (2)	Z (1)
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	1	0	0	0
6	1	0	0	1
7	1	0	1	0
8	1	0	1	1
9	1	1	0	0



Select one:

- $W \ xor \ Y$
- $W \ xnor \ Z$
- $W \ xnor \ Y$
- $W \ xor \ Z$ ✔

Your answer is correct.

The correct answer is: $W \ xor \ Z$

Question 19

Correct

Mark 1.00 out of 1.00

Convert a decimal number 6.625 to binary

6.625

Answer: 110.101 ✔

0.625×2 = 1 + 0.250
0.250×2 = 0 + 0.500
0.500×2 = 1

6 = 4+2+0
= 110.101

The correct answer is: 110.101

Question 20

Correct

Mark 1.00 out of 1.00

The below boolean expression has 4 minterms marked as m1,m2,m3 and m4.

$$F= \underbrace{a'bc'e}_{m1}+\underbrace{c'd'e'f}_{m2}+\underbrace{a'bc'd'f}_{m3}+\underbrace{a'bd'ef}_{m4}$$

Identify the redundant minterm.

Select one:

- m3 ✔
- m2
- m4
- m1

Your answer is correct.

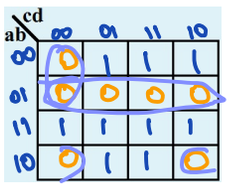
The correct answer is: m3

Question 21

Correct

Mark 1.00 out of 1.00

$F = ab + ad + b'd + a'b'c$ is populated in the K-map given below.



$F' = a'b + ab'd' + a'c'd'$

$F = (a+b')(a'+b+d)(a+c+d)$

0,4,5,6,7,8,10

Choose the correct maxterm representation.

Select one:

- $F(a,b,c,d) = [?][?]\Pi(0,4,5,6,7,8,11)$
- $F(a,b,c,d) = [?][?]\Pi(0,4,5,6,7,8,10)$ ✔
- $F(a,b,c,d) = [?][?]\Pi(1,2,3,9,10,12,13,$
- $F(a,b,c,d) = [?][?]\Pi(1,2,3,9,11,12,13,$

Your answer is correct.

The correct answer is:

$F(a,b,c,d) = [?][?]\Pi(0,4,5,6,7,8,10)$

Question 22
Correct
Mark 1.00 out of 1.00

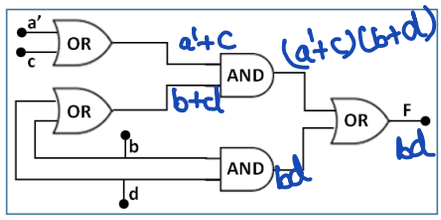
Convert a hexadecimal number $AB.CD$ to decimal number (correct to 3 decimal places)

Answer: 171.801

The correct answer is: 171.801

Question 23
Correct
Mark 1.00 out of 1.00

Boolean function F in the below circuit diagram is to be populated in the K-map shown alongside.



ab \ cd	00	01	11	10
00		1	1	
01	1	1	1	1
11		1	1	1
10			1	

Choose the correct minterm representation.

Select one:

- $F(a,b,c,d)=[?][?]\Sigma(1,2,4,5,8,9,10,12$
- $F(a,b,c,d)=[?][?]\Sigma(1,3,4,5,6,7,10,12$
- $F(a,b,c,d)=[?][?]\Sigma(1,3,4,5,6,7,11,13$ ✔
- $F(a,b,c,d)=[?][?]\Sigma(1,2,4,5,8,9,11,13$

Your answer is correct.

The correct answer is:
$F(a,b,c,d)=[?][?]\Sigma(1,3,4,5,6,7,11,13,1$

Question 24
Correct
Mark 1.00 out of 1.00

Convert a decimal number -10.375 to its 8-bit binary representation (including 1 sign bit as MSB)

Answer: 10101.101

The correct answer is: 10101.101

Question 25

Correct

Mark 1.00 out of 1.00

What is the maximum number of essential prime implicants in a 4 variable K-map

Answer: 8 ✔

The correct answer is: 8

Question 26

Incorrect

Mark 0.00 out of 1.00

The first step in minimizing a boolean function using the Quine-McCluskey method is to identify and segregate the minterms into groups. What is the maximum number of groups possible in the case of a 5 variable boolean function?

Answer: 16 ✖

The correct answer is: 6

◄ Mini Quiz 4 Jump to... MiniQuiz 5 (Part 1) ►